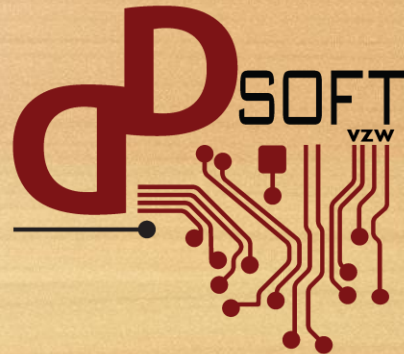


Making enabling apps for disabled people

Apps with data for people with a disability

with **Xamarin Forms**



Who am I?



Dennie Declercq.

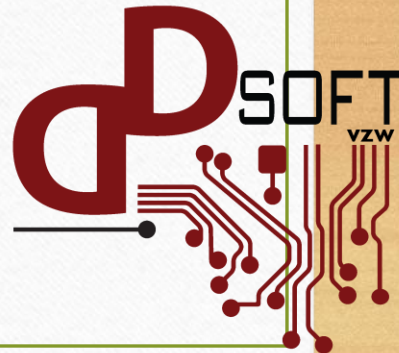
Founder of DDSOFT nonprofit.

Volunteer developer at Ithaka (a coaching center for people with an intellectual Disability).

Obsessed about using technologies to make people with a disability their life better.
Maybe because I have my own disability (autism).

Honored to speak here!

Inventor of the hashtag #DreamingIsBelieving



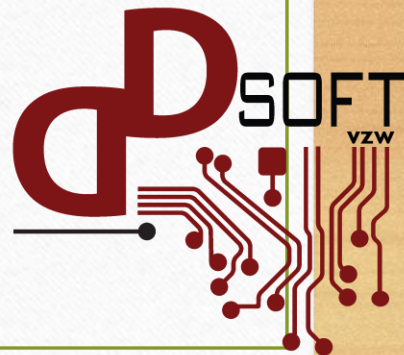
Technology we use

To show all the things, we create a **new Xamarin Forms** project.

For practical reasons, we build only for **Android** and **Windows 10** at this time!

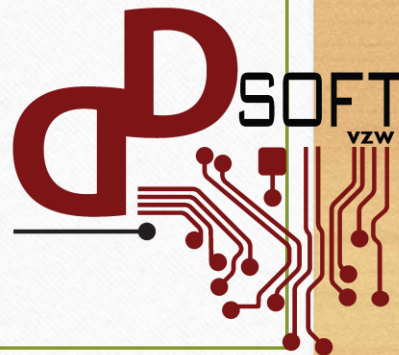


Xamarin



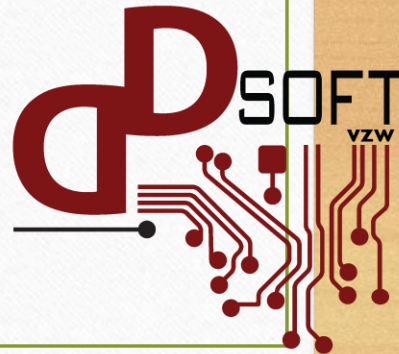
What this session is not about

- This session is not a deep dive into building apps in [Xamarin](#).
- I assume at least that you are common with creating basic apps in [Xamarin](#).



What this session is about

- This session is to let you know, how you can write apps for a new user-base with programming techniques we all know.



Different kinds of disabilities

Physical

- Can't move like the 'normal way'.

Mental / Intellectual

- Can't think / speak / read / behave (communicate) like the 'normal way'.

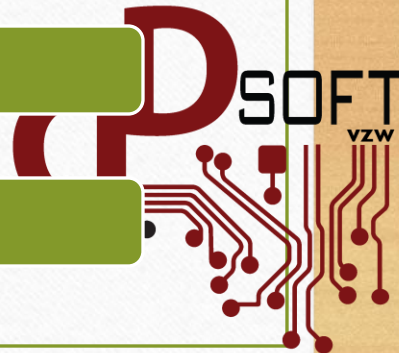
Visual

- They can see like 'the normal way'.

Audio

- They can't hear like 'the normal way'.

Other (specific and combinations)

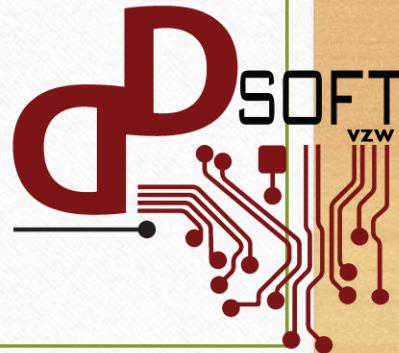


“

Eye openers

”

Let's start with a few “eye openers”.



Can you realize ?

*That the things that we're all do...
.. Are not common "life" for many people?*

Learning disability special

"Research has shown that with brain training exercises, children are able to learn and read faster, more easily and with greater comprehension," Du Plessis added.

Edublox, a South African company that helps children and adults with learning disabilities through specially developed programmes, have introduced a bursary programme for pupils with learning disabilities in less

fortunate communities. "Not being able to learn is debilitating and learners from these areas have a lot to contend with as it is," said Zainu Allie, Edublox franchise owner and facilitator in Cape Town. "We have been working closely with 16 learners from schools in the Hout Bay area since April. With the safe, fun and challenging environment provided, at our Centre in Claremont, we have seen

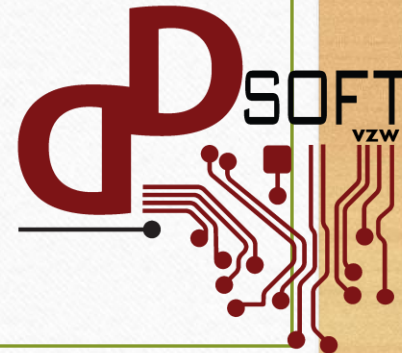


Reading a paper

Reading a subtitled movie



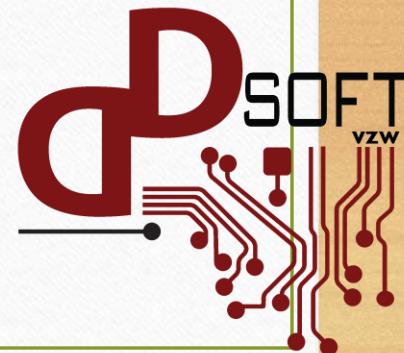
Ordering a pizza



What's this?



An eating
schedule



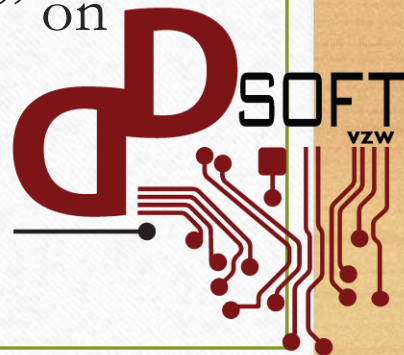
What's the purpose?

- To let somebody, know what's to eat this week.
- But what does that paper means,
- if you can't read?



There are **CA 2-3%** People with an intellectual disability in the world.

- What means **2 a 3 %** of the people cannot do everything like “we” on a cognitive / intellectual approach.



What barely exist?

There are (very less) tech organizations who focusses on people with physical disabilities

And that's good!

There are (less) tech organizations who focus on people with vision loss or hearing problems.

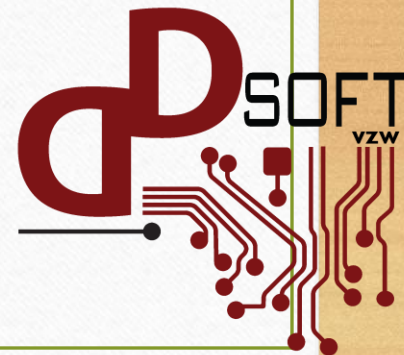
And that's also good!

But there are quit less tech organistions who focus on people with intellectual/ cognitive disabilities.

That's our (DDSoft nonprofit) goal.

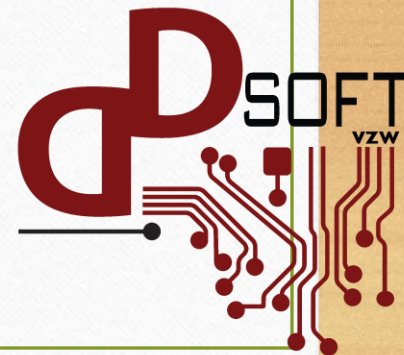
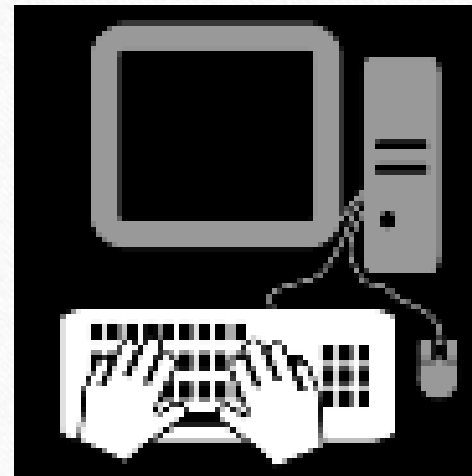
But I hope...

To inspire much more!

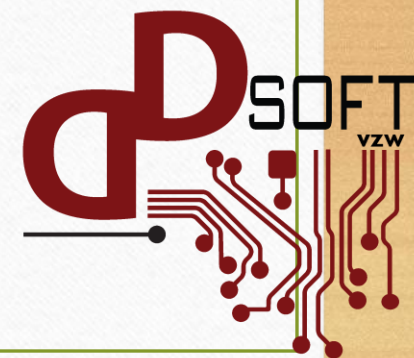
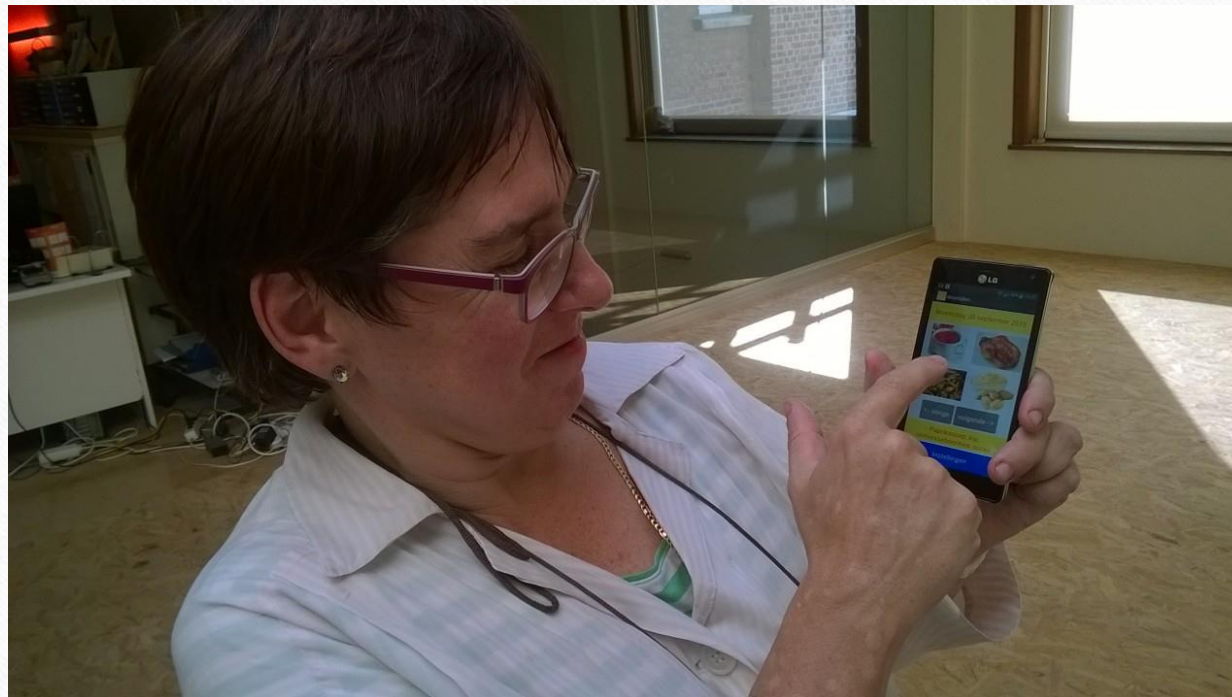


Where are developers for?

- For making others' life easier.
- To automate things that human cannot do;
- Or cannot do fast enough.
- I hope you are all here for this reason!

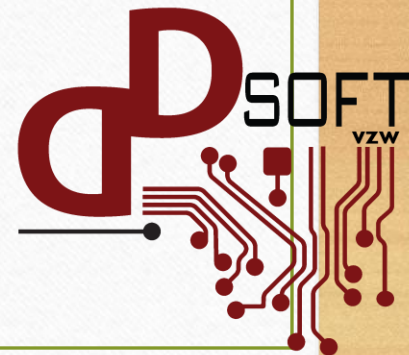


Let's learn to make apps for people with a disability!



Three basic rules

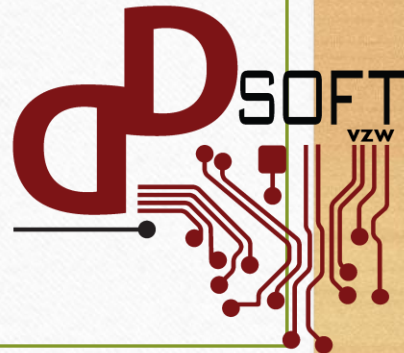
1. Simplify the layout;
2. Use pictographs or pictures;
3. Use Text-To-Speech;
4. (optional) An inclusive approach.



Demo Time 'Emoboard'!

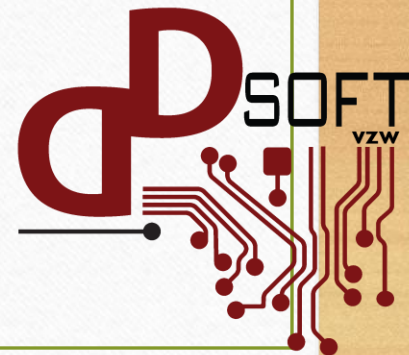
- Use case:
- Let's assume you want to build an app to help them communicate about their mood.

"Many people with a disability have difficulties to talk about their mood."



Demo Time – Starting point.

- We start this demo with an empty solution, with the **Xamarin Forms** template.
- We have the files `App.xaml` and `MainPage.xaml`.



The new Project Layout

New Cross Platform App - EmoBoard

Select a template:

 Blank App

 Master Detail

A cross-platform template for building cross-platform, native mobile apps for iOS, Android, and Universal Windows Platform.

Use the native user interface approach to create UIs for each platform, or use Xamarin.Forms to create cross-platform, native UIs in XAML.

UI Technology

☒ Xamarin.Forms

☐ Native

Code Sharing Strategy ?

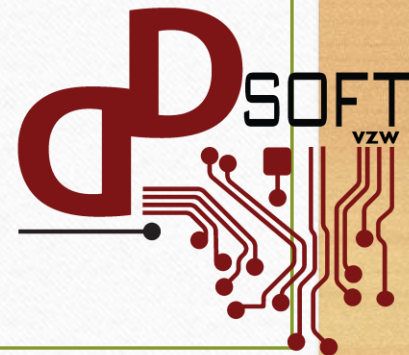
☐ Shared Project

☒ Portable Class Library (PCL)

Microsoft Azure

☐ Host in the cloud

OK Cancel



Simplify the layout

1. Use a simple white background color (nothings easier to read then white)
2. Use a few accent colors
3. Color by function
4. Be consistent!



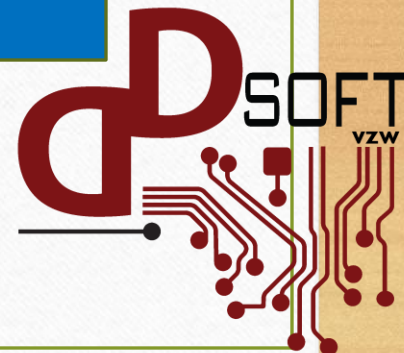
Color by function

I've created three button styles:

I can speak !

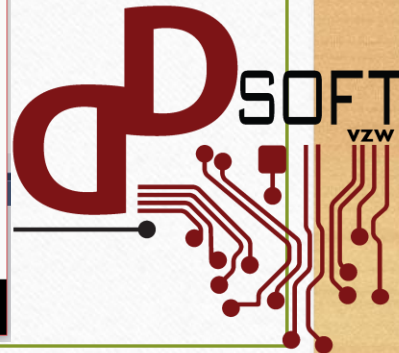
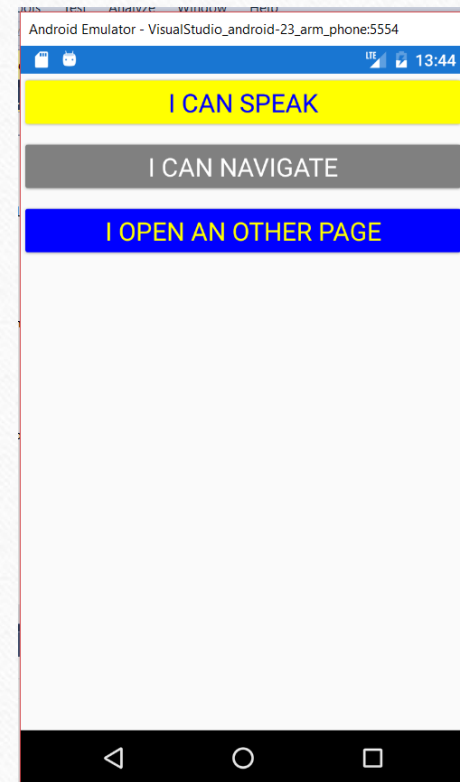
I can navigate !

I open an other page!



Color by funtion (in code)

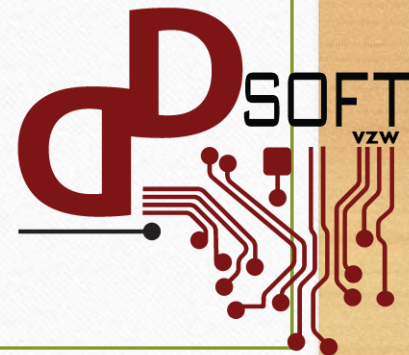
```
<StackLayout>
  <Button x:Name="SpeakButton"
    BackgroundColor="Yellow"
    TextColor="Blue"
    FontSize="Large"
    Text="I can speak"
    Clicked="SpeakButton_Clicked"/>
  <Button x:Name="NavigateButton"
    BackgroundColor="Gray"
    TextColor="White"
    FontSize="Large"
    Text="I can navigate"
    Clicked = "NavigateButton_Clicked"/>
  <Button x:Name="OtherPageButton"
    BackgroundColor="Blue"
    TextColor="Yellow"
    FontSize="Large"
    Text="I open an other page"
    Clicked = "OtherPageButton_Clicked"/>
</StackLayout>
```



Demo

Demo Time!

Making a low-barrier user interface for “Emoboard”.



Use pictographs or pictures

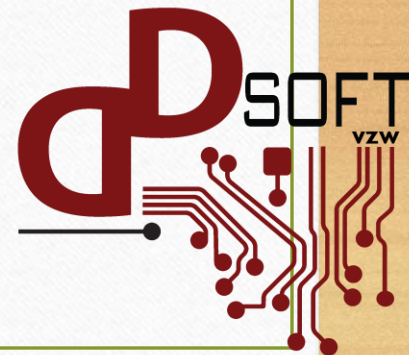
- What means the following?

- 酢飯

- The answer is:



For some people with a disability,
who cannot read,
Letters and words means nothing.
Just like Chinese or Japanese
tokens for us.
Everybody (except people with
visual impairments) can see a
picture.



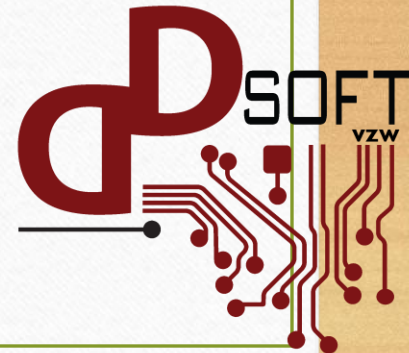
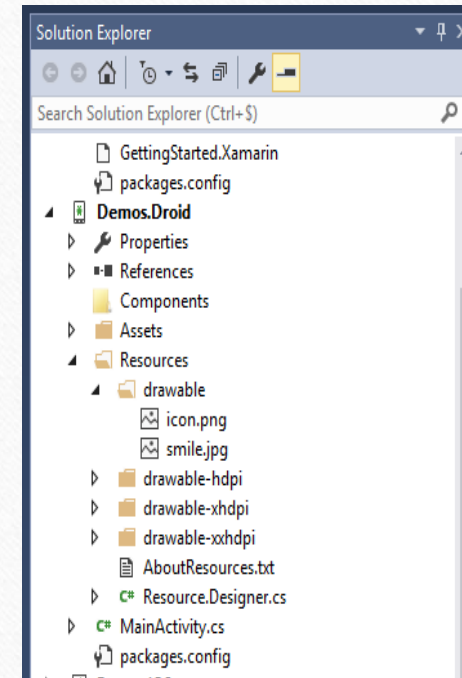
Use pictographs or pictures (in code)

```
ContentLayout="Top,0"
```

```
Image="Happy.png"
```

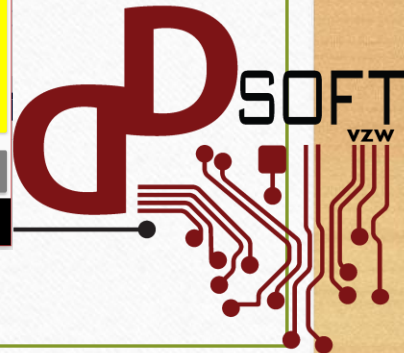
“ContentLayout” is needed to be sure that the image is on top and the text is below.

It's a good practice to show also the text!



Demo

- Demo Time!
- Adding some pictographs and pictures.



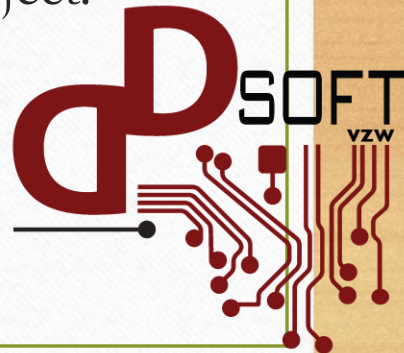
Use Text-To-Speech (TTS)

For some people, pictures aren't enough.

To reach them also, you can use **TTS, Text-To-Speech**

You have to do three steps:

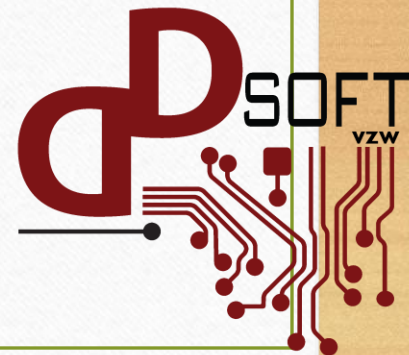
- 1 Add the nuget package CrossTextToSpeech by James Montemagno to **each** project.
- 2 Use just a simple line of code.
- 3 Modify your (emulator) device settings to enable TTS.



TTS Plugin – Simple code

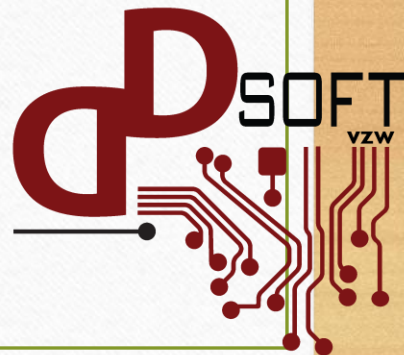
- → Add the speak function

```
using Plugin.TextToSpeech;  
private void HappyButton_Clicked(object sender, EventArgs e)  
{  
    CrossTextToSpeech.Current.Speak(HappyButton.Text.ToString());  
}
```



Demo

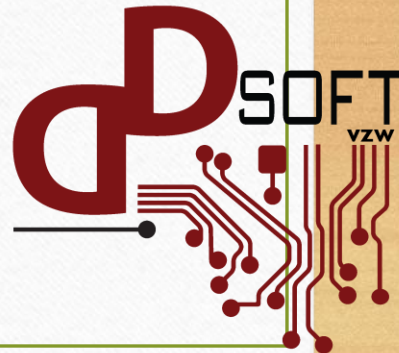
-
- Demo of Text-To-Speech



For people who doesn't need all te functions

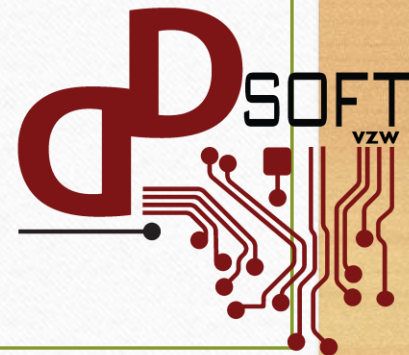
We want to make an **inclusive** app. This means that people with more needs have more assistance, and people with less need, less assistance.

To include this approach, we add a settings page where we can check the TTS function out.



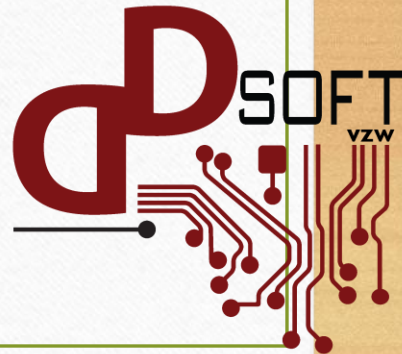
Adding the Settings Page

1. We adding the page SettingsPage.Xaml
2. We add a stackpanel whith the content vertical centred.
3. We add a textblock and a switch to enable or disable speech
4. A save and cancel button



Modyfing App.cs

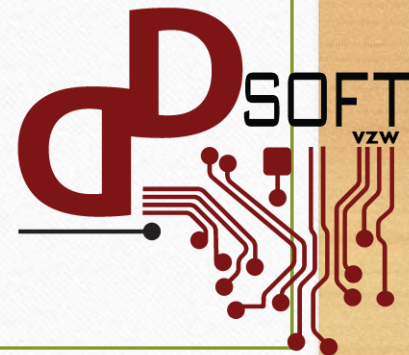
- We need to encapsulate MainPage.Xaml in a **NavigationPage**.
- It's important, otherwise you can't navigate to the SettingsPage



We add the settings.cs class

- The `settings.cs` class has the following code:

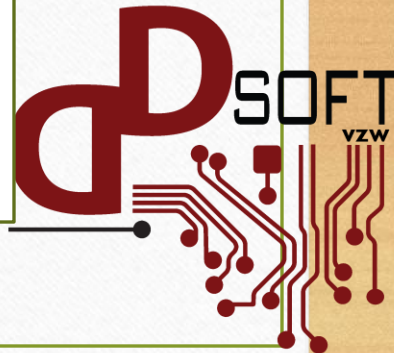
```
public static class Settings
{
    public static bool EnableTTS = true;
}
```



The code behind of SettingsPage.cs

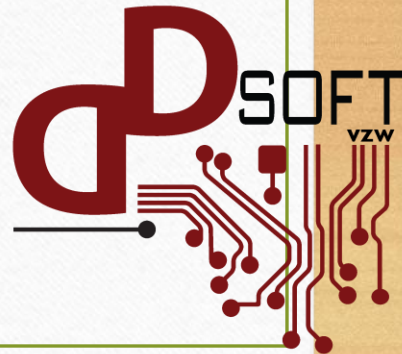
```
if(TTSToggle.IsToggled == true)
{
    Settings.EnableTTS = true;
} else
{
    Settings.EnableTTS = false;
}

await Navigation.PopToRootAsync();
```



Modifying our existing code.

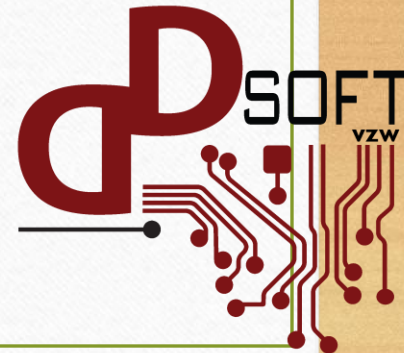
```
private void HappyButton_Clicked(object sender, EventArgs e)
{
    if(Settings.EnableTTS == true)
    {
        CrossTextToSpeech.Current.Speak(HappyButton.Text.ToString());
    }
}
```



Smoothing down the rough edges (SettingsPage.Xaml.cs)

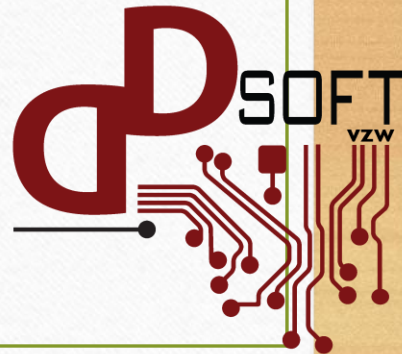
- Since we have a **Navigation Page**, we must set a **Title**.
- We must load the state of the **Toggle button**, otherwise it's always true when coming back on the page.

```
this.Title = "Settings";  
if(Settings.EnableTTS == true)  
{  
    TTSToggle.IsToggled = true;  
} else  
{  
    TTSToggle.IsToggled = false;  
}
```



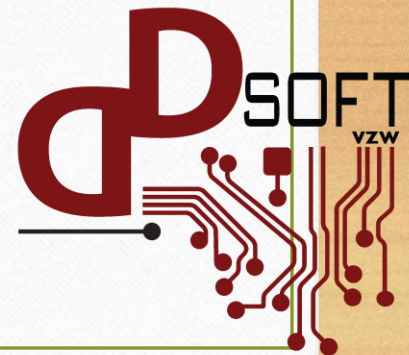
Smoothing the rough edges (MainPage.Xaml.cs)

- Also here, we have to make the page have a **title**.



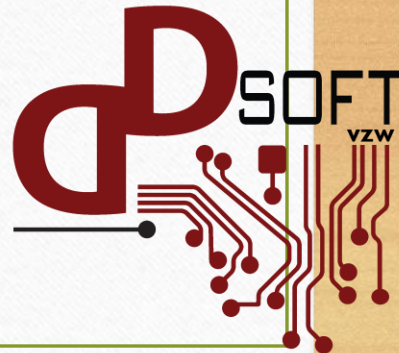
DEMO: Enable choiches (inclusive approach)

- Final demo



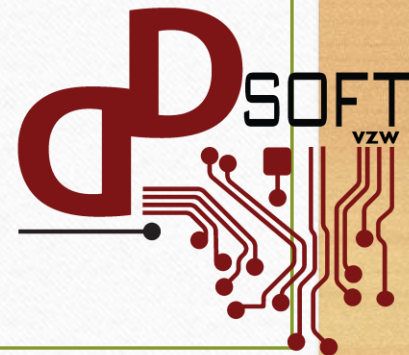
The last thing that isn't implemented

- If you make a production app, than you have to care that the settings must be saved in a database like **SQLite** or a **Settings Plugin**.
- We don't have time to do this today.



Recap

- Today you have seen how you can make an app accessible for people who cannot read or write due cognitive disabilities.
- You have learn to:
 1. Simplify the layout with color by function
 2. Use images and pictographs
 3. Use TTS
 4. Add an inclusive approach

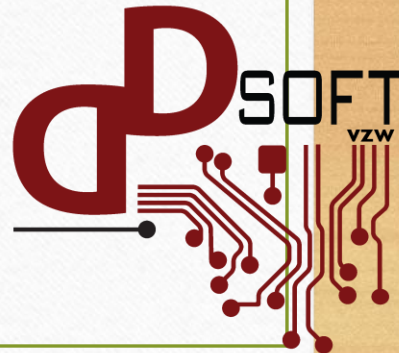


Now it's up to you!

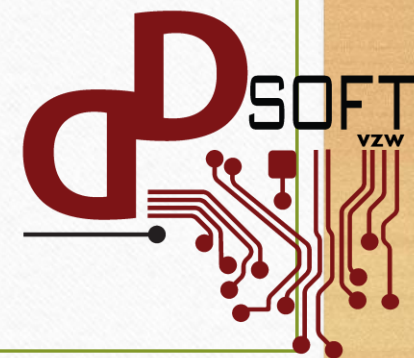
The journey of apps for people with a disability just starts here.

The next big thing is using data (like an open data platform)

So you cannot only let them explain their feelings, but also let them inform about their life.



Q&A



Thanks!

- I am Dennie Declercq.
- I live in Belgium.
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- You can call me: +32478294525.
- You can help me spread the word on Twitter:
[@DennieDeclercq](https://twitter.com/DennieDeclercq).

